

PROFILE	Affiliation: University of Oxford Email: richard.stiskalek@physics.ox.ac.uk NASA ADS: 18 first-author, 17 co-author papers; <i>h</i>-index = 13; citations = 886 Website: richard-sti.github.io I am a Career Development Research Fellow at St John’s College, Oxford, and a Hintze Prize Fellow in Astrophysics (Cosmology) at the University of Oxford. I study how structure formed in the nearby Universe, out to a few hundred megaparsecs, and how fast the Universe is expanding, specialising in Bayesian field-level inference and digital twins of the local Universe to bridge observations and theory. My research combines constrained cosmological simulations, semi-analytic galaxy formation models, simulation-based inference, and machine learning to test galaxy physics, map large-scale flows, and deliver precision measurements of cosmological parameters. I implement these as GPU-accelerated, differentiable models in Python and JAX, sampling millions of parameters on high-performance computing clusters.
EDUCATION	University of Oxford (Balliol College), DPhil in Astrophysics 2022 – 2026 Thesis: “Constrained Simulations of the Local Universe as a Laboratory for Precision Cosmology and Astrophysics” Supervised by <i>Harry Desmond</i>, <i>Julien Devriendt</i>, and <i>Adrianne Slyz</i> Ludwig-Maximilians-Universität München, MSc in Physics 2020 – 2022 Thesis: “Strong-field lensing of gravitational waves” (partially with <i>Albert Einstein Institute Potsdam</i>) Supervised by <i>Miguel Zumalacárregui</i>, <i>Marius A. Oancea</i>, and <i>Jochen Weller</i> Hong Kong University of Science and Technology 2017 – 2018 Undergraduate Exchange Programme in Physics University of Glasgow, BSc in Physics with Astrophysics 2016 – 2020 Thesis: “Gravitational-wave cosmology” Supervised by <i>Martin Hendry</i>
ACADEMIC APPOINTMENTS	University of Oxford, Hintze Centre for Astrophysical Surveys 2026 – 2029 Hintze Prize Fellow in Astrophysics (Cosmology) Flatiron Institute, Center for Computational Astrophysics 2025 Pre-Doctoral Fellow (with <i>Shy Genel</i> and <i>Lucia A. Perez</i>) Institut d’astrophysique de Paris 2024 Long Term Attachment (with <i>Guilhem Lavaux</i>) Max Planck Institute for Gravitational Physics, Observational Relativity and Cosmology 2020 Summer Research Intern (with <i>Collin Capano</i>) University of Oxford, Sub-department of Astrophysics 2019 Summer Research Intern (with <i>Harry Desmond</i>) University of Glasgow, Institute for Gravitational Research 2018 Summer Research Intern (with <i>John Veitch</i> and <i>Chris Messenger</i>)
COLLABORATION MEMBERSHIPS	Learning the Universe 2025 – present Simons Foundation collaboration developing next-generation cosmological simulations, machine learning methods, and advanced Bayesian inference for large-scale surveys <i>Contributions:</i> Training set generation and BORG team CAMELS 2025 – present Collaboration building large public suites of cosmological hydrodynamical and <i>N</i>-body simulations spanning broad cosmological and astrophysical parameter ranges to train and test machine learning methods <i>Contributions:</i> Leading projects on semi-analytic modelling and simulation-based inference Aquila Consortium 2022 – present

International collaboration developing Bayesian field-level inference techniques to reconstruct initial conditions and study large-scale structure formation
Contributions: Leading projects on data generation, benchmarking, large-scale flows and cosmological parameters

TEACHING,
MENTORING &
SERVICE

Student Supervision

<i>Noah Pierce</i> (BSc, Southampton; with H. Desmond & T. Yasin)	2025 – present
“Field-Level Inference of the Galaxy Luminosity Function”	
<i>Joshua Darne</i> (MPhys, Oxford; with T. Yasin & H. Desmond)	2024
“Radial acceleration relation in the NewHorizon hydrodynamical simulation”	
<i>Fedir Boreiko</i> (BSc, Manchester; with T. Yasin & H. Desmond)	2024
“The correlation between light and dark matter across cosmic time” [arXiv:2601.07799]	
<i>Enoch Ko</i> (BSc, Warwick; with T. Yasin & H. Desmond)	2024
“Dark matter and galaxy dynamics: enduring puzzles” [arXiv:2508.03569]	
<i>Catherine Gallagher</i> (MPhys, Oxford; with T. Yasin & H. Desmond)	2023 – 2024
“The influence of cosmic environment on galaxy properties” [arXiv:2503.14732]	
<i>James Harvey</i> (BSc, Oxford; with T. Yasin & H. Desmond)	2023 – 2024
“Machine learning the time of last major merger from spectroscopic data”	

Teaching Experience

Tutor, <i>MPhys C1 Astrophysics</i> , University of Oxford	2023 – 2024
Graduate tutorials in cosmology and stellar astrophysics	
Mentor, <i>Lumiere Education</i>	2023 – present
Supervision of senior high school students carrying out independent research projects	
Demonstrator, <i>3rd-Year Practical Course</i> , University of Oxford	2023
Computational astrophysics practical sessions	

Academic Service

Referee for <i>A&A</i> , <i>ApJ</i> , <i>MNRAS</i> , <i>OJA</i> , <i>PNAS</i> , and <i>PRD</i>	2022 – present
Reviewed 20+ manuscripts for leading journals	
Coordinator, Aquila Consortium Monthly Telecon	2023 – present
Organiser, “Simulator’s Coffee”, University of Oxford	2023 – 2024
Local organiser, Aquila Consortium Oxford Meeting	2023
Lead organiser, “Middle of Scotland Science Festival”	2018

SELECTED
AWARDS AND
SCHOLARSHIPS

Hintze Prize Fellowship in Astrophysics (Cosmology), University of Oxford	2026
Career Development Research Fellowship, St John’s College, Oxford	2026
Flatiron Research Fellowship, Flatiron Institute (declined)	2026
Junior Research Fellowship, Emmanuel College, Cambridge (declined)	2026
Snell Exhibition, Balliol College	2022 – 2026
STFC PhD Funding, Science and Technology Facilities Council	2022 – 2026
DAAD Study Scholarship, German Academic Exchange Service	2021 – 2022
Kerr Bursary, University of Glasgow	2020
Lang Scholarship, University of Glasgow	2019
Undergraduate Summer Bursary, Royal Astronomical Society	2018
Dean’s List, Hong Kong University of Science and Technology	2018
Astronomy 1 Prize, University of Glasgow	2017
Matthew A. Muir Bursary, University of Glasgow	2017
South East Asia Study Abroad Scholarship, University of Glasgow	2017 – 2018

SKILLS

Programming: Python, Julia, C/C++, Fortran; high-performance computing (MPI, GPU acceleration)
Statistics & Machine Learning: Bayesian inference (NumPyro/JAX, PyMC, Stan), simulation-based inference, density estimation; deep learning (MLPs, CNNs, GNNs, normalising flows)

with JAX, TensorFlow, PyTorch; tree-based methods (random forests, gradient boosting) with scikit-learn

Astrophysics Software: RAMSES, AREPO, Gadget, Rockstar, DisPerSe; N -body/hydrodynamical simulation analysis

Languages: English (fluent), Czech (native), Slovak (fluent), French (intermediate), German (beginner)

PUBLICATIONS **First-Author – Submitted**

- [1] “A reanalysis of the megamaser Hubble constant: from spot catalogues to peculiar velocities”; **R. Stiskalek**, H. Desmond [arXiv:2609.17684]

First-Author – Accepted

- [2] “Learning the Universe with cosmological rescaling of merger trees and semi-analytic galaxy formation models”; **R. Stiskalek**, L. A. Perez, S. Genel, R. S. Somerville, R. E. Angelo, S. Contreras [ApJ 1007:112, arXiv:2606.10024]
- [3] “Forward-modelling Milky Way Cepheids: selection effects and physical priors in the Gaia–HST calibration”; **R. Stiskalek**, A. Riess, H. Desmond, G. Lavaux, D. Scolnic [MNRAS 550:stag1272, arXiv:2603.09880]
- [4] “ S_8 from peculiar velocities: agreement with *Planck* for Tully–Fisher and supernovae, tension for the fundamental plane”; **R. Stiskalek** [MNRAS 550:stag1266, arXiv:2509.20235]
- [5] “Validating Digital Twins of the Local Universe with the Thermal Sunyaev-Zel’dovich Signal”; **R. Stiskalek**, H. Desmond [Open Journal of Astrophysics 9, arXiv:2601.15935]
- [6] “Revisiting the Great Attractor: The Local Group’s streamline trajectory, cosmic velocity and dynamical fate”; **R. Stiskalek**, H. Desmond, S. McAlpine, G. Lavaux, J. Jasche, M. J. Hudson [Open Journal of Astrophysics 9, arXiv:2601.08524]
- [7] “1.8 per cent measurement of H_0 from Cepheids alone”; **R. Stiskalek**, H. Desmond, E. Tsaprazi, A. Heavens, G. Lavaux, S. McAlpine, J. Jasche [MNRAS 546:staf2260, arXiv:2509.09665]
- [8] “No evidence for local H_0 anisotropy from Tully–Fisher or supernova distances”; **R. Stiskalek**, H. Desmond, G. Lavaux [MNRAS 546:staf2048, arXiv:2509.14997]
- [9] “The Velocity Field Olympics: Assessing velocity field reconstructions with direct distance tracers”; **R. Stiskalek**, H. Desmond, J. Devriendt, A. Slyz, G. Lavaux, M. J. Hudson, D. J. Bartlett, H. Courtois [MNRAS 545:staf1960, arXiv:2502.00121]
- [10] “Testing the local supervoid solution to the Hubble tension with direct distance tracers”; **R. Stiskalek**, H. Desmond, I. Banik [MNRAS 543:1556, arXiv:2506.10518]
- [11] “Symmetry in Hyper Suprime-Cam galaxy spin directions”; **R. Stiskalek**, H. Desmond [Res. Notes AAS 8 281, arXiv:2410.18884]
- [12] “Evaluating the variance of individual halo properties in constrained cosmological simulations”; **R. Stiskalek**, H. Desmond, J. Devriendt, A. Slyz [MNRAS 534:3120, arXiv:2310.20672]
- [13] “Probing general relativistic spin-orbit coupling with gravitational waves from hierarchical triple systems”; M. A. Oancea, **R. Stiskalek**, M. Zumalacárregui [MNRAS 535:L1, arXiv:2307.01903]
- [14] “On the fundamentality of the radial acceleration relation for late-type galaxy dynamics”; **R. Stiskalek**, H. Desmond [MNRAS 525:6130, arXiv:2305.19978]
- [15] “Frequency- and polarization-dependent lensing of gravitational waves in strong gravitational fields”; M. A. Oancea, **R. Stiskalek**, M. Zumalacárregui [Phys. Rev. D 109, 124045, arXiv:2209.06459]
- [16] “The scatter in the galaxy–halo connection: a machine learning analysis”; **R. Stiskalek**, D. J. Bartlett, H. Desmond, D. Anbajagane [MNRAS 514:4026, arXiv:2202.14006]

- [17] “The dependence of subhalo abundance matching on galaxy photometry and selection criteria”; **R. Stiskalek**, H. Desmond, T. Holvey, M. G. Jones [[MNRAS 506:3205](#), [arXiv:2101.02765](#)]
- [18] “Are stellar-mass binary black hole mergers isotropically distributed?”; **R. Stiskalek**, J. Veitch, C. Messenger [[MNRAS 501:970](#), [arXiv:2003.02919](#)]

Co-Author – Submitted

- [1] “Forward-modelling the Tolman and distance-duality tests with IllustrisTNG”; H. Desmond, T. Yasin, **R. Stiskalek**, S. von Hausegger [[arXiv:2606.27182](#)]
- [2] “Learning the Universe: Constrained simulations of the Coma galaxy cluster – I. Radial X-ray and Compton- y signatures”; U. P. Steinwandel, S. McAlpine, **R. Stiskalek**, Rüdiger Pakmor, V. Springel, E. Churazov, I. Khabibullin, J. Jasche, G. Lavaux, G. L. Bryan [[arXiv:2606.10028](#)]
- [3] “Testing cosmic anisotropy with cluster scaling relations”; T. Yasin, **R. Stiskalek**, H. Desmond, S. von Hausegger, P. G. Ferreira [[arXiv:2602.06007](#)]
- [4] “MEGATRON: Reproducing the Diversity of High-Redshift Galaxy Spectra with Cosmological Radiation Hydrodynamics Simulations”; H. Katz, . . . , **R. Stiskalek**, . . . [[arXiv:2510.05201](#)]
- [5] “MEGATRON: how the first stars create an iron metallicity plateau in the smallest dwarf galaxies”; M. Rey, . . . , **R. Stiskalek**, . . . [[arXiv:2510.05232](#)]
- [6] “MEGATRON: the impact of non-equilibrium effects and local radiation fields on the circumgalactic medium at cosmic noon”; C. Cadiou, . . . , **R. Stiskalek**, . . . [[arXiv:2510.05667](#)]

Co-Author – Accepted

- [7] “Introducing sapphire: Towards Hybrid Physics-Informed, Data-Driven Modeling of Galaxy Formation”; V. Pandya, . . . , **R. Stiskalek**, . . . [[ApJ](#), in press, [arXiv:2604.06318](#)]
- [8] “Parameterizing Dark Energy at the density level: A two-parameter alternative to CPL”; G. Montefalcone, **R. Stiskalek** [[PRD 114:043538](#), [arXiv:2603.25735](#)]
- [9] “Testing subhalo abundance matching with galaxy kinematics”; F. Boreiko, T. Yasin, H. Desmond, **R. Stiskalek**, M. J. Jarvis [[MNRAS](#), in press, [arXiv:2601.07799](#)]
- [10] “Constraining cosmological simulations with peculiar velocities: a forward-modeling approach”; A. Valade, N. Libeskind, D. Pomarède, **R. Stiskalek**, Y. Hoffman, S. Gottlöber, R. B. Tully [[A&A](#), in press, [arXiv:2602.03699](#)]
- [11] “The subtle statistics of the distance ladder: On the distance prior and selection effects”; H. Desmond, **R. Stiskalek**, J. A. Najera, I. Banik [[MNRAS:stag1144](#), [arXiv:2511.03394](#)]
- [12] “The galaxy-environment connection revealed by constrained simulations”; C. Gallagher, T. Yasin, **R. Stiskalek**, H. Desmond, M. J. Jarvis [[MNRAS 546:stag108](#), [arXiv:2503.14732](#)]
- [13] “Renzo’s rule revisited: A statistical study of galaxies’ baryon–dark matter coupling”; E. Ko, T. Yasin, H. Desmond, **R. Stiskalek**, M. J. Jarvis [[MNRAS 544:4288](#), [arXiv:2508.03569](#)]
- [14] “CosmoBench: A Multiscale, Multiview, Multitask Cosmology Benchmark for Geometric Deep Learning”; N. Huang, **R. Stiskalek**, J.-Y. Lee, A. E. Bayer, C. C. Margossian, C. K. Jespersen, L. A. Perez, L. K. Saul, F. Villaescusa-Navarro [[NeurIPS 2025](#), [arXiv:2507.03707](#)]
- [15] “The Manticore Project I: a digital twin of our cosmic neighbourhood from Bayesian field-level analysis”; S. McAlpine, J. Jasche, M. Ata, G. Lavaux, **R. Stiskalek**, C. S. Frenk, A. Jenkins [[MNRAS 540:716](#), [arXiv:2505.10682](#)]
- [16] “The CosmoVerse White Paper: Addressing observational tensions in cosmology with

systematics and fundamental physics”; Eleonora Di Valentino, . . . , **R. Stiskalek**,
. . . [Physics of the Dark Universe, Volume 49, 2025, 101965, arXiv:2504.01669]

- [17] “Inferring the Ionizing Photon Contributions of High-Redshift Galaxies to Reionization with JWST NIRCам Photometry”; N. Choustikov, **R. Stiskalek**, A. Saxena, H. Katz, J. Devriendt, A. Slyz [MNRAS 537:2273, arXiv:2405.09720]

INVITED TALKS	<i>Supernova-free local ladders for H_0</i> Hubble & BAO Tensions, ICG Portsmouth GDR CoPhy Episode 4, Clermont-Ferrand, France Cosmology seminar, UCL Cosmology Journal Club, University of Oxford <i>Bridging Machine Learning and Cosmology at Scale</i> PV Reconstruction online meeting <i>Cosmological Rescaling of Merger Trees for Semi-Analytic Galaxy Modelling</i> SBI for Galaxy Evolution, Cambridge CAMELS + Baryon Pasting Workshop, University of Osaka, Japan <i>Velocity Field Olympics</i> Leibniz Institute for Astrophysics Potsdam, Germany	2026
	<i>Forward modelling of peculiar velocities and large-scale structure</i> Institut d’Astrophysique de Paris <i>Pinning down the local Universe with digital twins</i> University of Cambridge Yale University <i>Do SH0ES Cepheids prefer the Planck Hubble constant?</i> National Astronomy Meeting, Durham <i>Towards accurate models and tests of the local Universe</i> Columbia University Princeton University Center for Computational Astrophysics, Flatiron Institute <i>Cosmological Rescaling of Merger Trees for Semi-Analytic Galaxy Modelling</i> Center for Computational Astrophysics, Flatiron Institute <i>Velocity Field Olympics</i> Center for Computational Astrophysics, Flatiron Institute Cosmic Flows 2025, Brisbane	2025
	<i>Velocity Field Olympics</i> University of Portsmouth	2024
	<i>Search for the optimal dark matter halo density profile</i> University of Oxford <i>Is the radial acceleration relation a fundamental correlation?</i> University of Oxford	2023
	<i>Frequency and polarization dependent propagation of gravitational waves</i> University of Glasgow LMU Munich Max Planck Institute for Gravitational Physics, Potsdam <i>The scatter in the galaxy–halo connection</i> Baryon Pasters Collaboration meeting	2022
	<i>The scatter in the galaxy–halo connection</i> LMU Munich	2021
	<i>Reversible-jump MCMC in gravitational-wave astronomy</i> Max Planck Institute for Gravitational Physics, Hannover <i>Are binary-black hole mergers isotropically distributed?</i> LIGO Scientific Collaboration Data Analysis telecon	2020
	<i>The relation between galaxies and dark matter halos</i> University of Oxford	2019